

AI Integration Report

Week 08 — AI Integration & Intelligent Features Development

GEMINI · HUGGINGFACE

Course	CSE 4104 — Software Development III
Team	CSE4104-7C-T05
Project Title	AI Academic Mentor
Instructor	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK
Section	7C
Submission Date	13 August 2026
GitHub Repository Link	GitHub - Priyo-1158/cse4104-7c-t05-ai-academic-mentor: AI Academic Mentor - Personalized Learning Assistant for Students · GitHub

Team Information

Role	Member Name	Student ID	Contribution
Team Leader	Shadman Sadequeen Priyo	11230121158	Project coordination, GitHub management, integration testing
Frontend Developer	Musabbir Hossain Chayon	11230121168	API integration, Report, Ulpages, Netlify deployment, Next.js
Backend Developer	Sk. Nadirul Haque Nadir	11230121155	Node.js, Express, routing, Express routes, MongoDB models, JWT authentication
AI Integration Lead	Shafin Kabir	11230121175	Gemini API, HuggingFace, AI routes

AI Workflow Design

Every AI feature follows the same seven-stage pipeline, shown below with the actual file each stage lives in. The frontend never talks to an AI provider directly — all calls are proxied through the Express backend so provider keys never reach the browser bundle.

1	User Input Student types a question, pastes notes, or sets quiz/plan parameters in a React page (Chatbot.jsx, Quiz.jsx, Summarizer.jsx, Planner.jsx).
2	Frontend Proxy Call aiApi.js (the GeminiAPI object) posts JSON to the backend — e.g. POST /api/ai/chat — never holding an AI key client-side.
3	Backend Route + Soft Auth The Express route in routes/ai.js runs softAuth middleware to attach req.user if a valid JWT is present, without blocking guest access.
4	Prompt Assembly processAIChat() combines a feature-specific system prompt with the user's input into one prompt string.
5	Provider Chain Gemini (gemini-2.5-flash) is tried first; on failure, HuggingFace (Qwen2.5-7B-Instruct); if both fail, a static template — the endpoint always responds 200.
6	Response Validation tryParseJSON() extracts/parses JSON for quiz/plan; chat/summary text is trimmed. The active source is tagged on every response.
7	Persist + Display Logged-in chat exchanges are written to MongoDB immediately; quiz/plan/summary results persist when the student saves them.

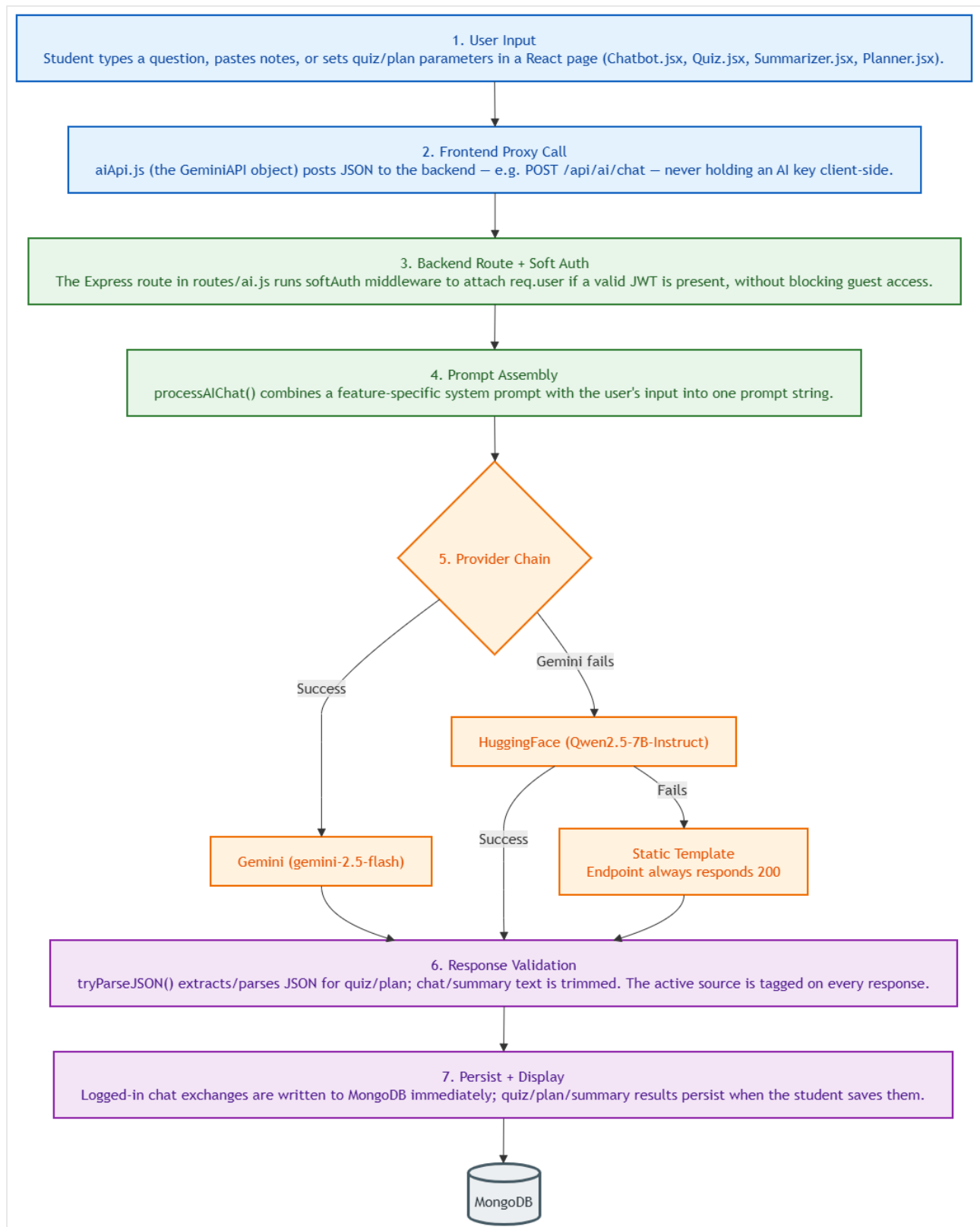


Figure : AI Workflow Design